

App Development Project Report

1. Student Details

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About me : I am a student of the IIT Madras BS Degree Programme(Data Science - Diploma level) with an interest in full-stack web development. I enjoy building real-world web applications that solve practical problems. This project helped me improve my understanding of Flask, SQLAlchemy, database design, authentication, and dashboard development.

2. Project Details

Project Title

Trek Management System

Problem Statement

To design and develop a web-based Trek Management System that allows administrators to manage treks, staff members to manage assigned treks, and users to browse, book, and cancel trek bookings through an easy-to-use interface.

Approach

The application was developed using Flask with SQLAlchemy ORM and SQLite database. The project follows a role-based access system consisting of Admin, Staff, and User modules.

The Admin manages users, staff approvals, treks, bookings, and dashboard analytics.

Staff members manage their assigned treks by updating trek status and available slots while monitoring active participants.

Users can register, log in, search available treks, make bookings, cancel bookings, and track their booking history through an interactive dashboard.

Interactive charts were implemented using Chart.js to visualize booking statistics and participation data.

3. AI / LLM Declaration

I used ChatGPT (GPT-5) to assist with:

- Debugging Flask routes
- SQLAlchemy query optimization
- Chart.js integration
- Bootstrap UI improvements
- Search and filter implementation
- Minor code refactoring and documentation

The overall AI usage was approximately **10–15%**.

All application logic, debugging, testing, integration, database design, and final implementation were completed manually.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend Web Framework
SQLAlchemy	ORM for SQLite Database
SQLite	Database
Jinja2	Dynamic HTML Templates
Bootstrap 5	Responsive User Interface
HTML5	Frontend Structure
CSS3	Styling
JavaScript	Client-side Interactions
Chart Js	Dashboard Charts
Werkzeug Security	Password Hashing

5. Database Schema / ER Diagram

User

- id
- username
- email
- password
- type
- is_approved
- is_blacklist

Trek

- id
- name
- location
- difficulty
- duration
- available_slot
- status
- staff_id

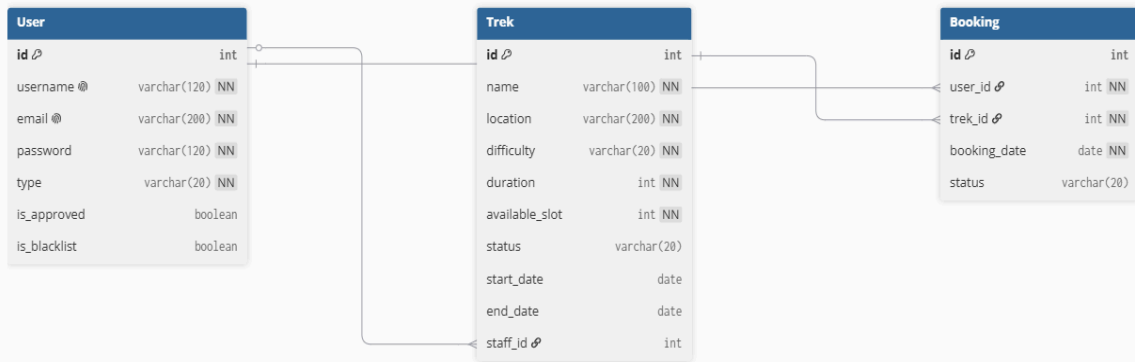
Booking

- id
- user_id
- trek_id
- booking_date
- status

Relationships

- One-to-Many → User → Booking
 - One user can make multiple trek bookings.
- One-to-Many → Trek → Booking
 - One trek can have multiple bookings from different users.

- One-to-Many → Staff → Trek
 - One staff member can manage multiple assigned treks.
- Many-to-One → Booking → User
 - Each booking belongs to one user.
- Many-to-One → Booking → Trek
 - Each booking belongs to one trek.
- Many-to-One → Trek → Staff
 - Each trek is assigned to one staff member.



6. API Resource Endpoints

Endpoint	Method	Description
/register	GET / POST	User Registration
/login	GET / POST	User Login
/logout	GET	Logout
/user_dashboard	GET	User Dashboard
/admin_dashboard	GET	Admin Dashboard
/staff_dashboard	GET	Staff Dashboard
/book_trek/<id>	GET	Book Trek
/cancel_booking/<id>	GET	Cancel Booking
/my_bookings	GET	Booking History
/admin/treks	GET	Manage Treks
/admin/users	GET	Manage Users
/admin/bookings	GET	Booking Records

7. Video Presentation

Drive Link:

 project_demo_23f3000214.mp4

<https://drive.google.com/file/d/18NopL3mKvUOr3iRK7AI92ofSh6jAUsub/view>